

# CUSTOMER SEGMENTATION ANALYSIS & BUSINESS STRATEGY RECOMMENDATIONS

**Subject:** Data-Driven Customer Segmentation for Enhanced Marketing ROI

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## 1. Executive Summary

This report presents a comprehensive customer segmentation analysis conducted on an e-commerce dataset comprising 2,000 customers. Utilizing advanced machine learning techniques—including K-Means clustering, Ridge regression, and Logistic regression—we have identified five distinct behavioural customer segments.

Our analysis reveals that **18.6% of customers (High-Value Loyal)** generate **50.6% of total revenue**, highlighting a critical dependency on a small, highly engaged group. Predictive models achieved an **R<sup>2</sup> score of 0.8334** for spending prediction and a **99.2% accuracy** for identifying future purchasers.

**Key Recommendation:** Reallocate 50% of the marketing budget toward "High-Value Loyal" and "New & Promising" segments, while initiating urgent re-engagement campaigns for the "At-Risk" segment to prevent revenue leakage.

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## 2. Business Problem & Objectives

### 2.1 Background

The e-commerce company faces challenges in allocating marketing spend efficiently due to a lack of distinct customer personas. Generic marketing campaigns result in wasted resources on low-engagement users and insufficient retention efforts for high-value patrons.

### 2.2 Objectives

- **Group Customers:** Segment customers based on purchasing behaviour (RFM - Recency, Frequency, Monetary).
  - **Predict Behaviour:** Forecast future spending and purchase likelihood using regression and classification models.
  - **Drive Action:** Provide actionable, segment-specific business strategies to maximize Customer Lifetime Value (CLV) and reduce churn.
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## 3. Data & Methodology

### 3.1 Dataset Overview

- **Sample Size:** 2,000 customers.
- **Features:** Recency (days since last purchase), Purchase Frequency, Total Spending, Average Order Value, Website Visits, Discount Usage Rate, Customer Rating, and Demographics.
- **Data Quality:** Zero missing values, outliers treated using the IQR (Interquartile Range) method.

### 3.2 Analytical Approach

- **Clustering (K-Means):** Applied Elbow Method and Silhouette Score (0.1934) to validate the optimal number of segments (k=5).
- **Regression (Ridge):** Predicted "Total Spending" using GridSearchCV for hyperparameter tuning (Alpha=1.0).

- **Classification (Logistic):** Predicted "Purchase Likelihood" with hyperparameter optimization (C=100).

## 4. Key Findings & Results

### 4.1 Customer Segments Profile

The following table summarizes the five distinct customer segments identified:

| Segment Name        | Customers (Count) | Revenue Share | Avg. Spending | Recency (Days) | Frequency | Discount Usage |
|---------------------|-------------------|---------------|---------------|----------------|-----------|----------------|
| 1. High-Value Loyal | 373               | 50.6%         | \$3,697       | 167            | 10.2      | 50%            |
| 2. New & Promising  | 438               | 15.7%         | \$976         | 186            | 8.3       | 50%            |
| 3. Discount-Driven  | 347               | 12.1%         | \$952         | 74             | 3.0       | 55%            |
| 4. At-Risk (Churn)  | 433               | 17.9%         | \$1,126       | 272            | 3.3       | 45%            |
| 5. Low-Engagement   | 409               | 3.6%          | \$243         | 186            | 2.5       | 51%            |
| Total/Average       | 2,000             | 100%          | \$1,362       | 182            | 5.5       | 50%            |

### 4.2 Predictive Model Performance

| Model Type          | Objective          | Best Parameters | Performance Metric   | Result |
|---------------------|--------------------|-----------------|----------------------|--------|
| Ridge Regression    | Predict Spending   | Alpha = 1.0     | R <sup>2</sup> Score | 0.8334 |
| Ridge Regression    | Predict Spending   | Alpha = 1.0     | RMSE                 | 513.72 |
| Logistic Regression | Predict Likelihood | C = 100         | Accuracy             | 99.2%  |
| Logistic Regression | Predict Likelihood | C = 100         | ROC-AUC              | 0.9999 |

**Key Insight:** *Purchase Frequency* is the strongest driver of total spending (Coefficient: +935), implying that efforts to increase repeat purchase rates directly amplify revenue.

## 5. Actionable Business Recommendations

Based on the distinct behavioural traits of each segment, the following strategic recommendations are proposed:

### 1. High-Value Loyal (373 Customers )

- **Strategy:** Implement a "Platinum VIP" loyalty program offering free expedited shipping, 24/7 priority support, and exclusive early access to new product drops.
- **Action:** Leverage personalized "thank you" campaigns and ask for user-generated content (reviews/photos).
- **Caution: Avoid heavy discounts.** These customers exhibit strong purchase intent without price incentives. Offering unnecessary discounts will only erode profit margins.

### 2. New & Promising (438 Customers )

- **Strategy:** Launch an automated "Onboarding Email Series" within the first 7 days of acquisition.
- **Action:** Provide a "**10% off the next order**" coupon to stimulate a second purchase. Recommend best-selling products based on their initial browsing category.
- **Goal:** Convert this group into loyal customers within the first 60 days to maximize long-term value.

### 3. Discount-Driven (347 Customers )

- **Strategy:** Capitalize on their price sensitivity by targeting them with limited-time "Flash Sales" and "Bundle Deals" (e.g., Buy 1 Get 1 50% off).
- **Action:** Use urgency tactics (countdown timers) and cart abandonment emails with a small incentive.
- **Caution:** Restrict discounting to specific campaign periods. Uncontrolled discounts will train these customers to wait for sales, harming regular-price revenue.

### 4. At-Risk (Churn) - *URGENT* (433 Customers)

- **Strategy:** Deploy an immediate "Re-engagement" campaign, as these customers have been inactive for an average of 272 days but possess high historical spending (\$1,126).
- **Action:** Send a "We miss you" email with a **15-20% comeback discount** or a free shipping code. Simultaneously, send a feedback survey ("How was your last experience?") to diagnose the root cause of churn.
- **Goal:** Recover these customers immediately to prevent permanent revenue loss.

### 5. Low-Engagement (409 Customers )

- **Strategy:** Minimize direct marketing spend on this segment.
  - **Action:** Route them to low-cost, automated email newsletters (monthly digests) featuring entry-level or budget-friendly products.
  - **Decision:** Reevaluate the cost-to-serve ratio. If they remain unresponsive, deprioritize them in favor of the four higher-value segments.
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## 6. Conclusion & Strategic Impact

**In summary**, our analysis successfully transitioned raw transactional data into a clear, actionable marketing strategy:

1. **Prioritize Retention:** Just 373 "High-Value Loyal" customers generate half of the revenue. Retention strategies for this group should be the highest priority.
2. **Budget Realignment:** We recommend shifting **50% of the marketing budget** to the **Loyal** and **New** segments, allocating the remaining resources to At-Risk recovery and Discount campaigns.
3. **Operational Deliverable:** A segmented customer list (customer\_segmentation\_final.csv) has been exported and is ready for direct integration into the CRM/marketing automation tool.

By implementing these data-backed strategies, the company can significantly improve customer retention rates, optimize marketing expenditure, and drive sustainable revenue growth.